Shrijani Vemulapally

Livermore, CA | (510) 931-0586 | sv554@cornell.edu | github.com/Shrijaniv | linkedin.com/in/shrijanivemulapally | Portfolio

Education

Cornell University August 2026 – May 2027
Master of Engineering, Computer Science

- **Relevant Coursework:** Robot Perception | Mobile Autonomous Robots (Spring 2027) | Frontiers of Computer Vision | Systems for Large-Scale Machine Learning | Systems Programming
- **Teaching Assistant, CS 2110 (Java):** Lead discussions and office hours; coordinate assignment grading and resolve rubric inconsistencies for a 300+ student course.

University of California, Irvine December 2024
B.S. Computer Science & Engineering — ***GPA: 3.8***

Projects & Research

SocialSense — Socially-Aware Robot | Research Assistant, Fussell Lab, Cornell August 2026 – Present

- Created a multimodal pipeline (yt-dlp, ffmpeg/OpenCV, Tesseract OCR, faster-whisper) that converts in-the-wild robot videos into a labeled dataset of social episodes.
- Generated VLM labels (Gemini/Claude) and counterfactual demonstrations, validated against human annotations with a human-label-first protocol and versioned prompts.
- Training an LLM policy on this data with exocentric-to-egocentric transfer so a robot can act proactively before being commanded.

LiDAR Sensor Characterization — Robot Perception (ECE 5240) October 2026

- Characterized a simulated planar LiDAR for stationary-target detection and localization, modeling first-hit ray geometry, fog attenuation (Beer–Lambert), and probabilistic returns in PyBullet.
- Validated detection with 1,000-trial Monte Carlo (Wilson intervals) and bounded localization precision via a Fisher-information/CRLB estimate; cross-checked the 2D model against PyBullet 3D ray casts to 0.04–0.06 m.

Semantic Segmentation (U-Net) — Frontiers of Computer Vision, Cornell October 2026

- Implemented a U-Net with a ResNet backbone in PyTorch for pixel-level segmentation; added class-weighted loss to correct foreground/background imbalance, reaching 95.9% mean per-class accuracy on the binary task.

Pixory — iOS Photo Storytelling App | GitHub June 2026 – Present

- Shipped an iOS app (React Native/Expo, Node.js/Fastify, Python vision service) that turns 300 trip photos into a captioned, sequenced story; in private beta with six users.
- Designed a selection funnel that clusters photos by time and location, scores 120 candidates on nine visual signals (OpenCV, InsightFace), and has GPT-4o sequence the top 30.
- Personalized selection with a feedback loop that re-weights scoring from each user's swaps and edits, adapting to their taste within five uses.

Relevant Experience

Software Development Engineer, Amazon February 2025 – August 2026
Health Services, Growth and Marketing Engagement (GAME) *Seattle, WA*

- Architected an asynchronous EventBridge, Lambda, and Bedrock pipeline serving LLM inference in production at 6,200 peak RPS, partnering with the PM from requirements through deployment.
- Built LLM-as-judge evaluation that scores generated model output for safety and quality before delivery.
- Engineered internal AI agents with MCP tool integrations over a code/docs knowledge base to automate engineering workflows end-to-end.

Software Development Engineer Intern, Amazon June 2024 – September 2024
Health Services *Seattle, WA*

- Developed a RAG prototype for an Amazon Health team of 10–15 Product Managers to explore customer-feedback insights through natural-language queries.
- Automated S3-triggered ingestion in SageMaker: chunked data, generated Titan embeddings, and stored them in DocumentDB for KNN retrieval and Claude 3.5 Sonnet answers.

Technical Skills

Languages: Python | C++ | Java | TypeScript | Rust

Robotics & Perception: PyBullet | LiDAR & Sensor Modeling | Semantic Segmentation | State Estimation (CRLB) | Computer Vision

AI & ML: PyTorch | LLMs | VLMs | RAG | Data Pipelines | LLM Evaluation | Embeddings | Agentic Workflows

Tools: OpenCV | ffmpeg | Whisper (ASR) | AWS | SageMaker | NumPy / pandas | Git